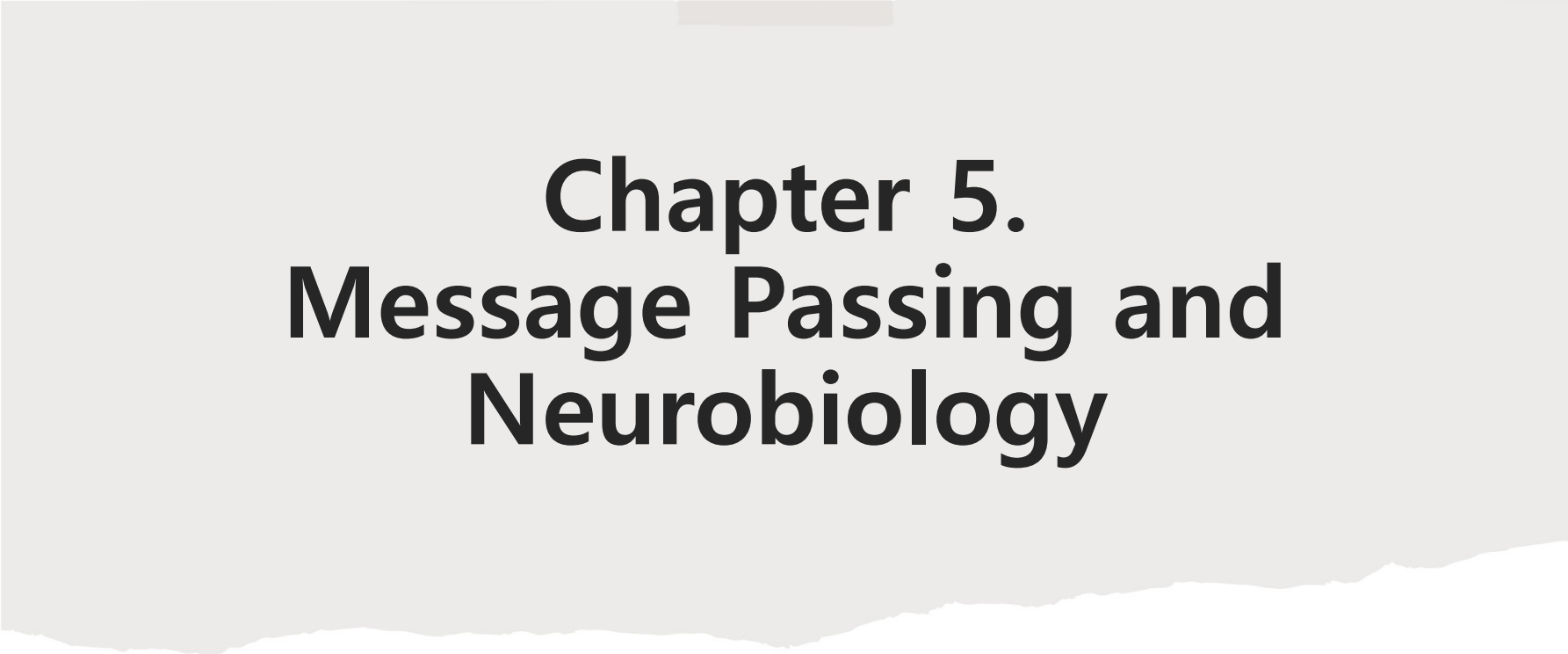




Chapter 5. Message Passing and Neurobiology

Active Inference: The Free Energy Principle in Mind, Brain, and Behavior

이서현 (건국대)

Introduction: Principles vs. Process Theories

- **The Principle (Ch. 1-3)**
 - What serves as the normative goal?
 - → Minimization of **Variational Free Energy**
- **The Process Theory (Ch. 5)**
 - How is this implemented in the brain?
 - From mathematical equations to neuronal dynamics
- **Key Premise**
 - **Belief Updating $\approx$ Neuronal Firing Rates**
 - **Message Passing $\approx$ Synaptic Transmission**

The Canonical Microcircuit

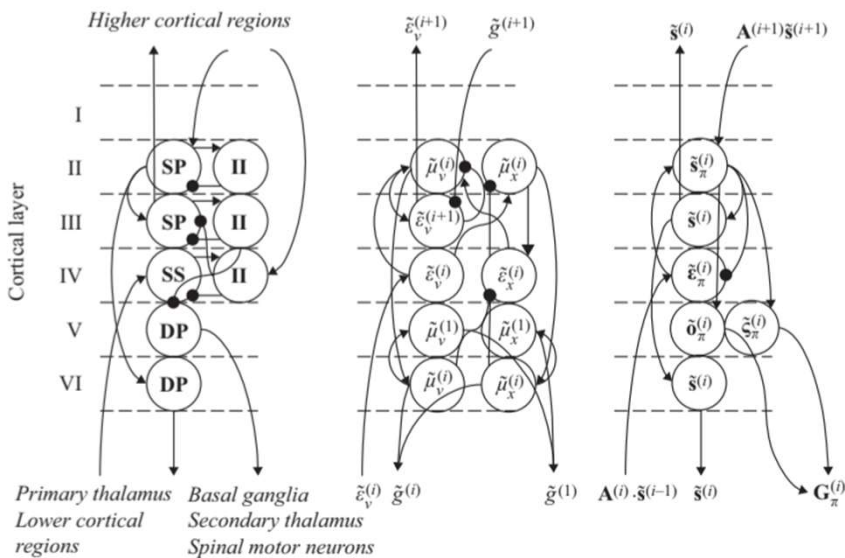


Figure 5.1

Thomas Parr et. al. (2022), Figure 5.1

- The Architecture of Inference
 - The cortex exhibits a stereotyped laminar structure (6 layers)
 - Active Inference maps **Hierarchical Predictive Coding** onto this anatomy
- Computational schema (middle) vs. biological microcircuit (left)
 - **Prediction Errors (ϵ)**: Ascending signals (Supragranular layers)
 - **Predictions / Expectations (μ)**: Descending signals (Infragranular layers)

- Message Passing

- **Ascending vs. Descending Streams**

- **Forward Pass (Bottom-up):** Carries **Prediction Errors**

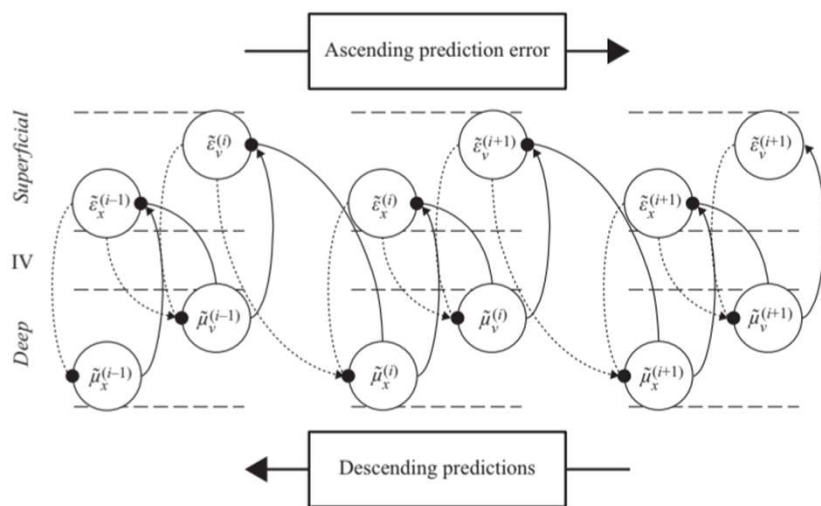
- Origin: Superficial Pyramidal Cells (Layers II/III)
 - Target: Layer IV (Spiny Stellate Cells)

- **Backward Pass (Top-down):** Carries **Predictions**

- Origin: Deep Pyramidal Cells (Layers V/VI)
 - Target: Deep & Superficial layers

- **Spectral Asymmetry**

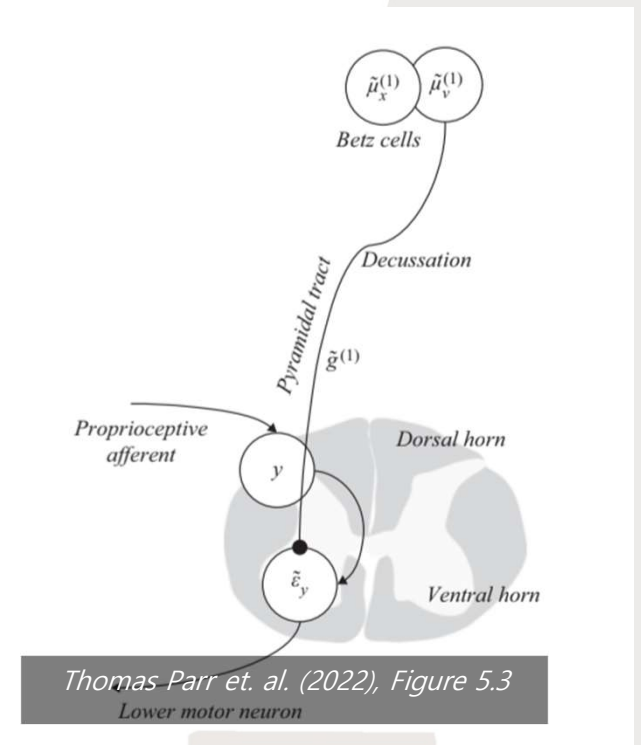
- **Prediction Errors:** Associated with fast frequencies (e.g., **Gamma**)
 - **Predictions:** Associated with slower frequencies (e.g., **Alpha/Beta**)



Thomas Parr et. al. (2022), Figure 5.2

Motor Control: Action as Inference

- **The Problem of Action**
 - How does the brain cause muscles to move?
- **The Active Inference Solution**
 - Action is not a "command" but a **fulfillment of a prediction**.
 - The brain predicts: "My limb is moving."
 - This generates a **Proprioceptive Prediction Error** at the spinal level.
 - Reflex arcs resolve this error by contracting muscles



Sensory Attenuation

- **The Condition for Movement**
 - If we trust our current sensory input (that the limb is static), we cannot move
 - We must **attenuate** the precision of ascending sensory prediction errors
 - **"Ignoring the reality to realize the belief"**
- **Clinical Implications**
 - **Sensory Attenuation:** Necessary for movement initiation
 - **Failures:**
 - Too much attenuation → Hallucinations?
 - Too little attenuation → Akinesia (Parkinson's), Catatonia

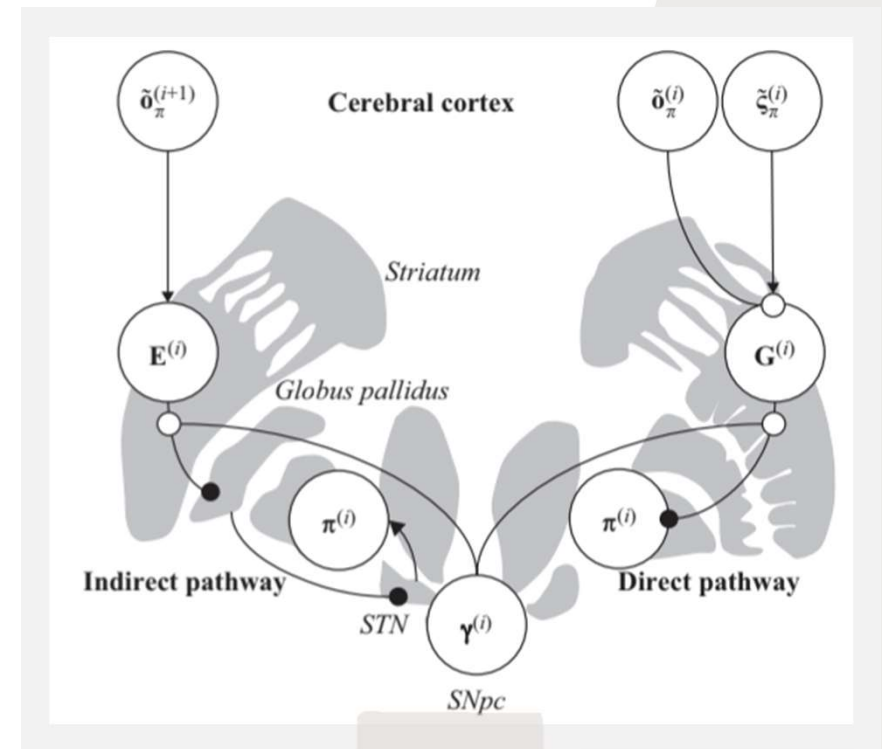
Subcortical Structures: The Basal Ganglia

- **Decision Making & Policy Selection**

- Cortex computes states; Basal Ganglia selects **Policies** (sequences of actions)

- **Anatomy of Choice**

- **Direct Pathway:** Facilitates selected policies (Go)
- **Indirect Pathway:** Suppresses competing policies (No-Go)



The Role of Dopamine

- **Dopamine as Precision**
 - In Active Inference, Dopamine $\neq$ Reward
 - **Dopamine $\approx$ Precision over Policies**
 - It encodes the **confidence** that a selected policy will minimize Expected Free Energy
- **Mechanism**
 - High Dopamine $\rightarrow$ High precision $\rightarrow$ **Selection (Action)**
 - Low Dopamin $\rightarrow$ Low precision $\rightarrow$ **Uncertainty / Inaction**
 - Explains tremor and akinesia in Parkinson's disease (loss of dopaminergic cells)

Synaptic Plasticity and Learning

- **Learning in the Brain**
 - Inference (States) = Fast dynamics (Firing rates)
 - Learning (Parameters) = Slow dynamics (Synaptic weights)
- **Minimizing Free Energy**
 - Optimizing model parameters to reduce Free Energy leads to **Hebbian-like** rules
 - **Associative Plasticity:** "Neurons that fire together, wire together"
 - This is mathematically derived from the gradient descent on Free Energy

Conclusion: The Biologically Plausible Brain

- **Summary**
 - **Cortical Microcircuits** implement Hierarchical Predictive Coding
 - **Reflex Arcs** implement Action through Proprioceptive Predictions
 - **Neuromodulators (Dopamine)** encode Precision (Confidence)
- **Takeaway**
 - Active Inference offers a unified **Process Theory** that bridges the gap between abstract computational principles and the tangible wetware of neurobiology

Discussion Points

The Paradox of Sensory Attenuation Context

"To move, we must attenuate sensory evidence."

If ignoring reality is a prerequisite for action, how do we distinguish between "healthy movement" and "hallucination"? Does "facing reality" hinder action?

Rethinking Dopamine Context

Dopamine encodes Precision over Policies, not Reward.

How does this reshape our understanding of Addiction or Motivation? Can we have action without reward, driven solely by high precision (confidence)?

Biological Plausibility vs. Implementation Context

The specific mapping of layers and frequencies (Fig 5.1)

Is this specific architecture the *only* way to implement Active Inference? What if neurobiological evidence contradicts the Gamma/Beta frequency hypothesis?
